

MATH 3551 A — Abstract Algebra I

Additional Practice Midterm 1 — Version A

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For practice — Midterm 1 is Monday, October 12, 2026 • Time limit: 50 minutes • Total: 100 points

NAME (please print): _____

Directions. This exam is **closed book**: no textbook, no notes, no calculators, no phones, no other devices, and no assistance of any kind. Write your answers on the exam itself; use the backs of pages if you need more room, and say clearly when you have done so. *Show your work and name the theorems you use* — “by Lagrange’s Theorem, $|H|$ divides 24” earns nearly all of the credit, an unjustified correct number earns little. Point values are marked on each problem. Good luck.

Version A was generated with GPT-5.6 Luna in Codex. Please report any error, ambiguity, out-of-coverage question, or timing issue to Professor Dupuy.

Problem 1. (30 points) **True or false; prove or disprove.**

Circle **T** or **F**, and then give a one-line reason: a theorem, a short computation, or an explicit counterexample. Five points each: two for the circle, three for the reason.

- (a) **T** **F** There is an element of order 12 in the symmetric group S_7 .
- (b) **T** **F** The map $\varphi: \mathbb{Z}/8\mathbb{Z} \rightarrow \mathbb{Z}/8\mathbb{Z}$ defined by $\varphi(\bar{n}) = \overline{2n+1}$ is a group homomorphism.
- (c) **T** **F** The congruence $8x \equiv 6 \pmod{14}$ has a solution.
- (d) **T** **F** The set $H = \{\bar{0}, \bar{3}, \bar{6}, \bar{9}\}$ is a subgroup of the additive group $\mathbb{Z}/12\mathbb{Z}$.
- (e) **T** **F** The number of transpositions in S_8 is 32.
- (f) **T** **F** The group $(\mathbb{Z}/12\mathbb{Z})^\times$ is cyclic.

Problem 2. (25 points) **Arithmetic modulo 91 and 63.**

- (a) (6 points) Use the Euclidean algorithm to find integers u, v such that $37u + 91v = 1$.
- (b) (5 points) Find the multiplicative inverse of $\overline{37}$ in $\mathbb{Z}/91\mathbb{Z}$, and solve $37x \equiv 14 \pmod{91}$.
- (c) (8 points) Solve $21x \equiv 42 \pmod{63}$. Describe all solution classes without listing them one at a time.
- (d) (6 points) How many elements does the group $(\mathbb{Z}/91\mathbb{Z})^\times$ have? Justify your answer.

Problem 3. (25 points) **A matrix subgroup.**

In $\text{GL}_2(\mathbb{R})$, let

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad H = \langle A, B \rangle.$$

- (a) (4 points) Compute A^2 and A^4 , and determine $|A|$.
- (b) (6 points) Compute BAB^{-1} and $(AB)^2$.
- (c) (10 points) Prove that every element of H is one of A^i or A^iB for $0 \leq i < 4$, and list these eight matrices. Prove that they are distinct.
- (d) (5 points) Is H cyclic? Justify your answer by comparing AB and BA .

Problem 4. (20 points) **Subgroups of $\mathbb{Z}_2 \times \mathbb{Z}_4$.**

Let $G = \mathbb{Z}_2 \times \mathbb{Z}_4$, with componentwise addition.

- (a) (4 points) Find the orders of $(1, 0)$, $(0, 1)$, $(1, 1)$, and $(1, 2)$.
- (b) (6 points) List the elements of each of $\langle(1, 1)\rangle$, $\langle(1, 2)\rangle$, and $\langle(0, 1)\rangle$.
- (c) (8 points) Let $K = \{(0, 0), (1, 0), (0, 2), (1, 2)\}$. Prove that K is a subgroup of G , and determine $K \cap \langle(1, 1)\rangle$ and $K \cap \langle(0, 1)\rangle$.
- (d) (2 points) Is G cyclic? Give a reason.